

4.0 SERIAL PERIPHERAL INTERFACE (SPI)

4.1 Overview

The ENC28J60 is designed to interface directly with the Serial Peripheral Interface (SPI) port available on many microcontrollers. The implementation used on this device supports SPI mode 0,0 only. In addition, the SPI port requires that SCK be at Idle in a low state; selectable clock polarity is not supported.

Commands and data are sent to the device via the SI pin, with data being clocked in on the rising edge of SCK. Data is driven out by the ENC28J60 on the SO line, on the falling edge of SCK. The $\overline{\text{CS}}$ pin must be held low while any operation is performed and returned high when finished.

FIGURE 4-1: SPI INPUT TIMING

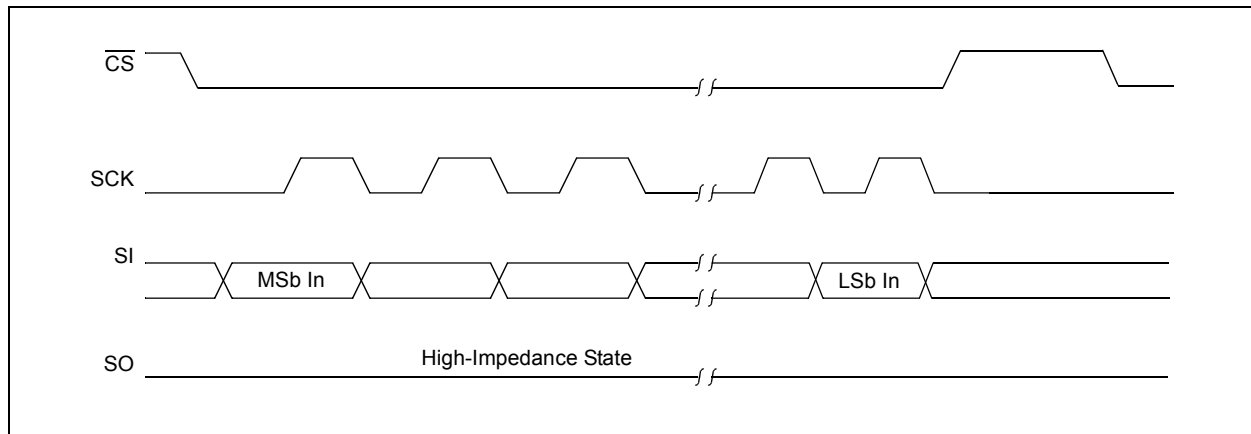
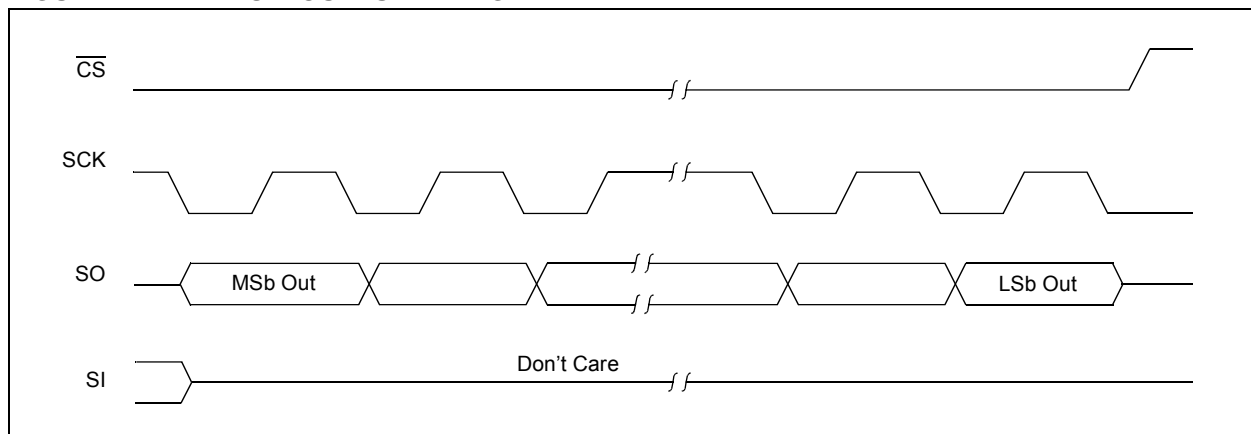


FIGURE 4-2: SPI OUTPUT TIMING



ENC28J60

4.2 SPI Instruction Set

The operation of the ENC28J60 depends entirely on commands given by an external host controller over the SPI interface. These commands take the form of instructions, of one or more bytes, which are used to access the control memory and Ethernet buffer spaces. At the least, instructions consist of a 3-bit opcode,

followed by a 5-bit argument that specifies either a register address or a data constant. Write and bit field instructions are also followed by one or more bytes of data.

A total of seven instructions are implemented on the ENC28J60. Table 4-1 shows the command codes for all operations.

TABLE 4-1: SPI INSTRUCTION SET FOR THE ENC28J60

Instruction Name and Mnemonic	Byte 0		Byte 1 and Following
	Opcode	Argument	Data
Read Control Register (RCR)	0 0 0	a a a a a	N/A
Read Buffer Memory (RBM)	0 0 1	1 1 0 1 0	N/A
Write Control Register (WCR)	0 1 0	a a a a a	d d d d d d d d
Write Buffer Memory (WBM)	0 1 1	1 1 0 1 0	d d d d d d d d
Bit Field Set (BFS)	1 0 0	a a a a a	d d d d d d d d
Bit Field Clear (BFC)	1 0 1	a a a a a	d d d d d d d d
System Reset Command (Soft Reset) (SRC)	1 1 1	1 1 1 1 1	N/A

Legend: a = control register address, d = data payload.

4.2.1 READ CONTROL REGISTER COMMAND

The Read Control Register (RCR) command allows the host controller to read any of the ETH, MAC and MII registers in any order. The contents of the PHY registers are read via a special MII register interface (see **Section 3.3.1 “Reading PHY Registers”** for more information).

The RCR command is started by pulling the $\overline{\text{CS}}$ pin low. The RCR opcode is then sent to the ENC28J60, followed by a 5-bit register address (A4 through A0). The 5-bit address identifies any of the 32 control

registers in the current bank. If the 5-bit address is an ETH register, then data in the selected register will immediately start shifting out MSb first on the SO pin. Figure 4-3 shows the read sequence for these registers.

If the address specifies one of the MAC or MII registers, a dummy byte will first be shifted out on the SO pin. After the dummy byte, the data will be shifted out MSb first on the SO pin. The RCR operation is terminated by raising the $\overline{\text{CS}}$ pin. Figure 4-4 shows the read sequence for MAC and MII registers.

FIGURE 4-3: READ CONTROL REGISTER COMMAND SEQUENCE (ETH REGISTERS)

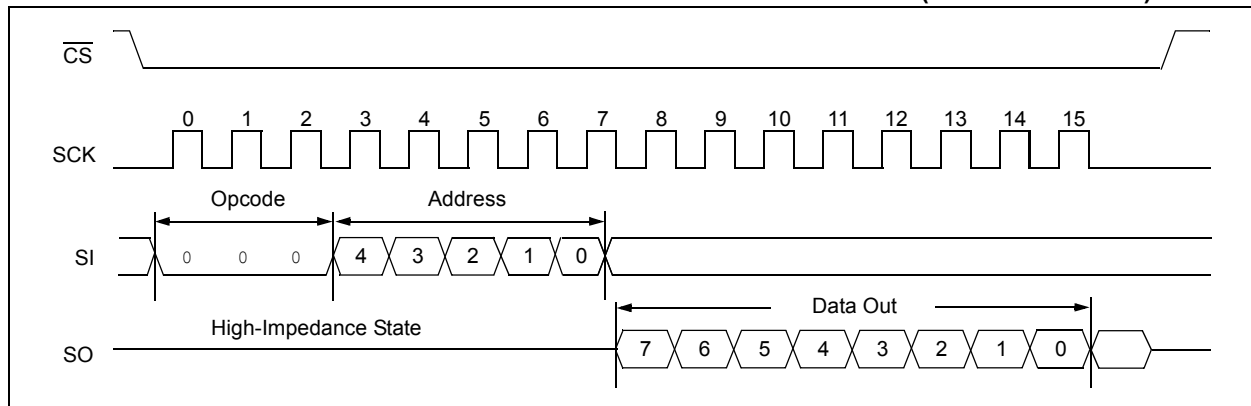
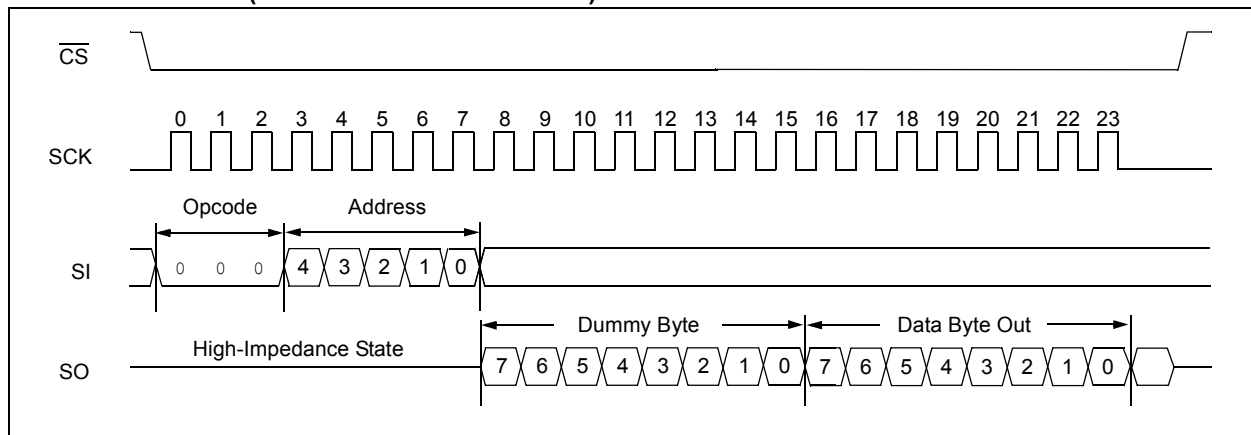


FIGURE 4-4: READ CONTROL REGISTER COMMAND SEQUENCE (MAC AND MII REGISTERS)



4.2.2 READ BUFFER MEMORY COMMAND

The Read Buffer Memory (RBM) command allows the host controller to read bytes from the integrated 8-Kbyte transmit and receive buffer memory.

If the AUTOINC bit in the ECON2 register is set, the ERDPT Pointer will automatically increment to point to the next address after the last bit of each byte is read. The next address will normally be the current address incremented by one. However, if the last byte in the receive buffer is read (ERDPT = ERXND), the ERDPT Pointer will change to the beginning of the receive buffer (ERXST). This allows the host controller to read packets from the receive buffer in a continuous stream without keeping track of when a wraparound is needed. If AUTOINC is set when address, 1FFFh, is read and ERXND does not point to this address, the Read Pointer will increment and wrap around to 0000h.

The RBM command is started by pulling the $\overline{CS}$ pin low. The RBM opcode is then sent to the ENC28J60, followed by the 5-bit constant, 1Ah. After the RBM command and constant are sent, the data stored in the memory pointed to by ERDPT will be shifted out MSb first on the SO pin. If the host controller continues to provide clocks on the SCK pin, without raising $\overline{CS}$, the byte pointed to by ERDPT will again be shifted out MSb first on the SO pin. In this manner, with AUTOINC enabled, it is possible to continuously read sequential bytes from the buffer memory without any extra SPI command overhead. The RBM command is terminated by raising the CS pin.

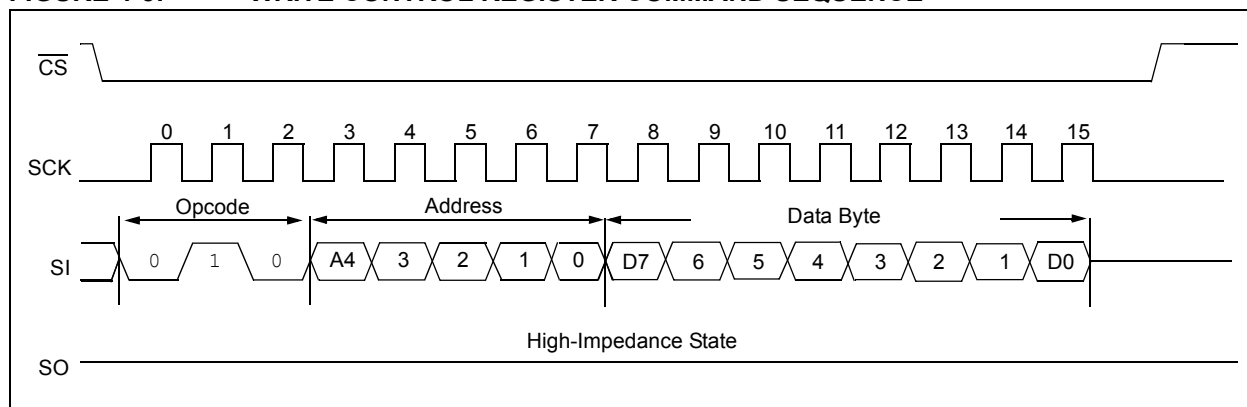
4.2.3 WRITE CONTROL REGISTER COMMAND

The Write Control Register (WCR) command allows the host controller to write to any of the ETH, MAC and MII Control registers in any order. The PHY registers are written to via a special MII register interface (see **Section 3.3.2 “Writing PHY Registers”** for more information).

The WCR command is started by pulling the $\overline{CS}$ pin low. The WCR opcode is then sent to the ENC28J60, followed by a 5-bit address (A4 through A0). The 5-bit address identifies any of the 32 control registers in the current bank. After the WCR command and address are sent, actual data that is to be written is sent, MSb first. The data will be written to the addressed register on the rising edge of the SCK line.

The WCR operation is terminated by raising the $\overline{CS}$ pin. If the $\overline{CS}$ line is allowed to go high before eight bits are loaded, the write will be aborted for that data byte. Refer to the timing diagram in Figure 4-5 for a more detailed illustration of the byte write sequence.

FIGURE 4-5: WRITE CONTROL REGISTER COMMAND SEQUENCE



4.2.4 WRITE BUFFER MEMORY COMMAND

The Write Buffer Memory (WBM) command allows the host controller to write bytes to the integrated 8-Kbyte transmit and receive buffer memory.

If the AUTOINC bit in the ECON2 register is set, after the last bit of each byte is written, the EWRPT Pointer will automatically be incremented to point to the next sequential address (current address + 1). If address, 1FFFh, is written with AUTOINC set, the Write Pointer will increment to 0000h.

The WBM command is started by lowering the $\overline{CS}$ pin. The WBM opcode should then be sent to the ENC28J60, followed by the 5-bit constant, 1Ah. After the WBM command and constant are sent, the data to be stored in the memory pointed to by EWRPT should be shifted out MSb first to the ENC28J60. After 8 data bits are received, the Write Pointer will automatically increment if AUTOINC is set. The host controller can continue to provide clocks on the SCK pin and send data on the SI pin, without raising $\overline{CS}$, to keep writing to the memory. In this manner, with AUTOINC enabled, it is possible to continuously write sequential bytes to the buffer memory without any extra SPI command overhead.

The WBM command is terminated by bringing up the $\overline{CS}$ pin. Refer to Figure 4-6 for a detailed illustration of the write sequence.

4.2.5 BIT FIELD SET COMMAND

The Bit Field Set (BFS) command is used to set up to 8 bits in any of the ETH Control registers. Note that this command cannot be used on the MAC registers, MII registers, PHY registers or buffer memory. The BFS command uses the provided data byte to perform a bit-wise OR operation on the addressed register contents.

The BFS command is started by pulling the $\overline{CS}$ pin low. The BFS opcode is then sent, followed by a 5-bit address (A4 through A0). The 5-bit address identifies any of the ETH registers in the current bank. After the BFS command and address are sent, the data byte containing the bit field set information should be sent, MSb first. The supplied data will be logically ORed to the content of the addressed register on the rising edge of the SCK line for the D0 bit.

If the $\overline{CS}$ line is brought high before eight bits are loaded, the operation will be aborted for that data byte. The BFS operation is terminated by raising the $\overline{CS}$ pin.

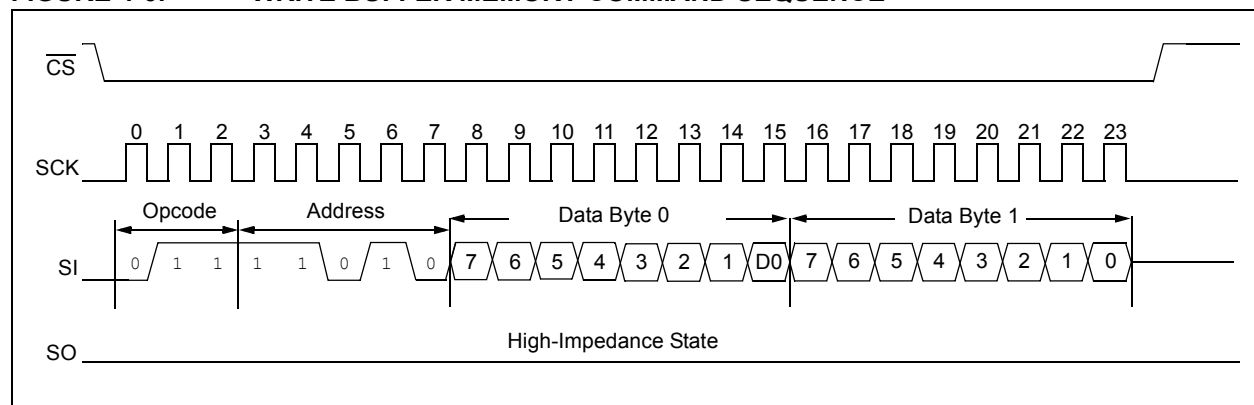
4.2.6 BIT FIELD CLEAR COMMAND

The Bit Field Clear (BFC) command is used to clear up to 8 bits in any of the ETH Control registers. Note that this command cannot be used on the MAC registers, MII registers, PHY registers or buffer memory. The BFC command uses the provided data byte to perform a bit-wise NOTAND operation on the addressed register contents. As an example, if a register had the contents of F1h and the BFC command was executed with an operand of 17h, then the register would be changed to have the contents of E0h.

The BFC command is started by lowering the $\overline{CS}$ pin. The BFC opcode should then be sent, followed by a 5-bit address (A4 through A0). The 5-bit address identifies any of the ETH registers in the current bank. After the BFC command and address are sent, a data byte containing the bit field clear information should be sent, MSb first. The supplied data will be logically inverted and subsequently ANDed to the contents of the addressed register on the rising edge of the SCK line for the D0 bit.

The BFC operation is terminated by bringing the $\overline{CS}$ pin high. If $\overline{CS}$ is brought high before eight bits are loaded, the operation will be aborted for that data byte.

FIGURE 4-6: WRITE BUFFER MEMORY COMMAND SEQUENCE



4.2.7 SYSTEM RESET COMMAND

The System Reset Command (SRC) allows the host controller to issue a System Soft Reset command. Unlike other SPI commands, the SRC is only a single byte command and does not operate on any register.

The command is started by pulling the $\overline{CS}$ pin low. The SRC opcode is sent, followed by a 5-bit Soft Reset command constant of 1Fh. The SRC operation is terminated by raising the $\overline{CS}$ pin.

Figure 4-7 shows a detailed illustration of the System Reset Command sequence. For more information on SRC's Soft Reset, refer to **Section 11.2 “System Reset”**.

FIGURE 4-7: SYSTEM RESET COMMAND SEQUENCE

